Principles of Microeconomics
Module 1: Thinking Like an Economist

Seeun Park

We begin with a basic but important question: **"What is economics?"** Next, we'll look at five key ideas that form the foundation for how economists think. Finally, we'll see how economists build models, which are usually simple but powerful tools for understanding the world.

By the end of this module, you'll be able to:

1. **Define** economics and **distinguish** between microeconomics and macroeconomics.
2. **Identify and describe** the five foundations of economics using real-world examples.
3. **Explain** why economists use models and assumptions, and **differentiate** between positive and normative analysis.

Icebreakers to Test Poll Everywhere

▶ [Poll]

1 Welcome to the World of Economics

What Is Economics?

Economics is not just about money or businesses! It's about understanding how people, firms, and governments make choices on nearly everything.

- **Economics** is the study of how individuals and societies allocate limited resources to satisfy unlimited wants.
- **Microeconomics** focuses on how individuals, households, firms, and governments make decisions in areas such as labor, health, education, finance, crime, and even personal choices like marriage.
- **Macroeconomics** examines the overall performance and structure of an economy, including topics like inflation, economic growth, unemployment, interest rates, and productivity.

☐ Economists try to explain so many different things in the world! (Activity1)

2 The Five Foundations of Economics

Economics is built on five key principles that help us understand how people make decisions. These ideas may seem simple at first, but they form the foundation for how economists think.

2.1 Incentives

People respond to incentives such as rewards or punishments that influence behavior. Incentives help explain why individuals, businesses, and governments make the choices they do.

- **Positive incentive:** encourage action.
  Example: Offering a gift card to people who donate blood increases participation.
- **Negative incentives** discourage action.
  Example: Losing points for turning in an assignment late pushes you to meet the deadline.

2.2 Trade-offs

Because resources are limited, we can't have everything we want. Making one choice means giving up something else.

- *Example:* Choosing to stay up late watching YouTube means giving up sleep. The more videos you watch, the less rest you get. That's a trade-off.

2.3 Opportunity Cost

The opportunity cost is the value of the next best alternative you give up when you make a decision. Thinking about opportunity cost helps us make better, more informed choices.

- *Example:* If you choose to go to bed, your opportunity cost is the value you could have gotten from watching YouTube.

2.4 Marginal Thinking

Marginal thinking means considering the additional (or marginal) benefit and cost of a decision. You might think the word "marginal" sounds unfamiliar, but in fact, we think at the margin all the time, often without even realizing it.

- *Example:* After eating one ice cream cone, you think about whether or not to order a second. You compare the price of the second cone with the extra happiness (or satisfaction) it might bring.

2.5 Trade Creates Value

Trade allows people to specialize in what they do best and exchange goods and services with others. Voluntary trade creates value and makes everyone involved better off.

- *Example:* You don't like cooking but don't mind washing dishes. Your friend loves to cook. By trading tasks, with your friend cooking and you doing the dishes, both of you are happier than if you each did everything yourself.

☐ Let's Practice! (Activity2)

3 How Economists Build Models

Economists use simplified models to make sense of complex real-world phenomena. These models focus on key relationships and rely on assumptions to highlight what matters most.

- Economic models are not meant to be perfectly realistic. They simplify the world to highlight key relationships.
- Like a map, a good model leaves out unnecessary details so we can focus on what matters most.
- Models are based on assumptions, and their value comes from how useful they are, not how detailed or "real" they are.

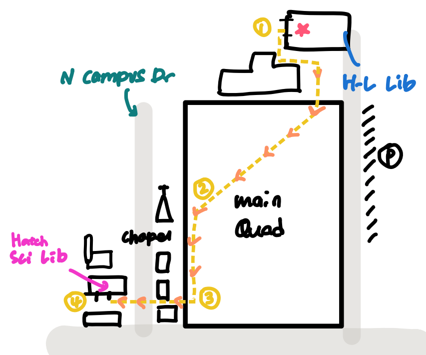


Figure 1: Far from realistic, but hopefully helpful.

Positive vs. Normative Statements

As economists, we often begin by describing how the world works, not how we wish it would work. In graduate school, especially as an applied microeconomist, I often heard the phrase: "Let the data speak." The idea is to step back from opinions and let models and evidence guide the conclusions.

- **Positive statements** are descriptive. They are claims about how the world *is* and can be tested or validated using data.
  - *Example:* "An increase in the minimum wage may reduce teen employment."
- **Normative statements** are prescriptive. They express a judgment about how the world *should* be and reflect values or priorities. Unlike positive statements, normative statements *cannot be tested or validated* using data.
  - *Example:* "The government should increase the minimum wage to reduce inequality."

☐ Let's Practice! (Activity2)

Practice Section

Activity 1: Micro vs. Macro — You Decide! (Topic 1)

Below are examples of real research topics in economics. Can you guess whether each one falls under microeconomics or macroeconomics?

RESEARCH HIGHLIGHTS ARTICLE

The downside of great expectations

January 14, 2022

How do overly optimistic growth forecasts affect a nation's economy?

RESEARCH HIGHLIGHTS ARTICLE

Few companies, fewer products

June 5, 2020

The oligopolistic smartphone market isn't producing enough phones for consumers.

RESEARCH HIGHLIGHTS ARTICLE

Teacher diversity in the classroom

January 10, 2023

Does exposure to same-race teachers help students in the long run?

RESEARCH HIGHLIGHTS ARTICLE

The uneven labor market impact of industrial robots

March 6, 2025

How are the effects of automation spread across the population?

RESEARCH HIGHLIGHTS FEATURED CHART

The ripple effects of officer injuries

June 5, 2023

Are police officers more likely to use force after one of their peers has been injured on duty?

RESEARCH HIGHLIGHTS ARTICLE

The spillover effects of trade wars

July 31, 2020

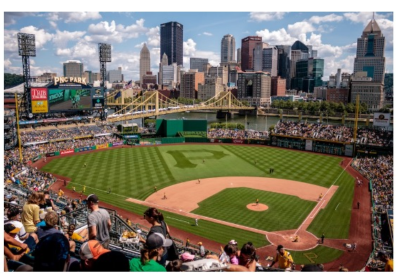
The 2018 tariffs caused price spikes to products that weren't even protected.

RESEARCH HIGHLIGHTS FEATURED CHART

Thank-you calls and donor retention

April 24, 2023

Does calling donors to thank them for their contributions increase the likelihood of future donations?



RESEARCH HIGHLIGHTS FEATURED CHART

Beating the heat

January 29, 2025

How much are people willing to pay to avoid uncomfortable temperatures?

Source: The American Economic Association's Research Highlights

Activity 2: What Economic Concept Is This? (Topic 2 and Topic 3)

Match each scenario below to one of the following concepts:

(A) Positive Incentives **(B)** Negative Incentives **(C)** Trade-offs

(D) Opportunity Cost **(E)** Marginal Thinking **(F)** Trade Creates Value

(G) Positive statement **(H)** Normative statement

▶ [Poll]

1. A bakery considers baking one more batch of cookies. The owner compares the extra cost of ingredients and labor to the revenue from selling the batch.
2. Some cities offer High-Occupancy Vehicle (HOV) lanes, which are special freeway lanes reserved for cars with two or more passengers.
3. A software engineer leaves a high-paying tech job in the city to open a coffee shop in their small hometown. What do we call the salary and lifestyle they gave up?
4. California has the perfect weather for growing oranges. Maine is known for catching the best lobster. If each state specializes in what it does best and they trade, both benefit.
5. You have two exams tomorrow morning, back to back. You only have a few hours left to study tonight. If you focus more on one subject, you'll be less prepared for the other.
6. New York City charges a fee for vehicles entering high-traffic areas under its Congestion Pricing Program.
7. The government should offer universal health insurance so that everyone can access the care they need.
8. Offering health insurance increases utilization of health care.